The University of the South Pacific

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School of Computing, Information and Mathematical Sciences

CS211: Computer Organisation

Test 1 – Semester 1, 2014

Time Allowed: 1 hour

Total Marks: 40 (10% of final grade)

Solution

Name: _____ **Student ID:** _____

INSTRUCTIONS:

- There are 8 questions in this paper and all are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

Question 1

a) What is the largest positive number that can be represented in Wombat 1 machine? Represent the number in bits and decimal as well. (2 marks)

0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 (1)

$2^{15} - 1 = 32767$ (1)

b) What is the smallest negative number that can be represented in Wombat 1 machine? Represent the number in bits and decimal as well. (2 marks)

1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 (1)

$= -2^{15} = -32768$ (1)

c) Find the sum of the decimal values 23 and -9 in 8 bit binary using two's complement? (2 marks)

23 ⇒ 0 0 0 1 0 1 1 1

\+ -9 + 1 1 1 0 0 1 1 1

14 1 0 0 0 0 1 1 1 0

overflow - Discard.

d) A 32-bit word on a little-endian computer has the decimal value of 261. If it is transmitted to a big-endian computer byte by byte and stored there (with byte 0 in byte 0 and so on), what is its decimal value on the big-endian machine? (3 marks)

261 in 32 bits ⇒ $\underbrace{0\cdots01}_{\text{Byte 1}}\ \underbrace{00000101}_{\text{Byte 0}}$

Little end:

∴ $\underbrace{0\cdots0}_{B3}\ \underbrace{0\cdots0}_{B2}\ \underbrace{00000001}_{B1}\ \underbrace{00000101}_{B0}$ (1)

Big endian: $\underbrace{00000101}_{B0}\ \underbrace{00000001}_{B1}\ \underbrace{0\cdots0}_{B2}\ \underbrace{0\cdots0}_{B3}$ (1)

∴ 00000101 00000001 00 ····· 0 = 8395161

Question 2

Convert -20.3 to IEEE single-precision format. Represent the final answer in hexadecimal. (4 marks)

-20.3 ⇒ IEEE, Base = 2, Excess = 127, sign bit = 1

20 ⇒ 10100

0.3

× 2

0 .6

× 2

1 .2

× 2

0 .4

× 2

0 .8

× 2

1 .6

× 2

1 .2

× 2

0 .4

∴ 20.3 = 10100.01001100110011...

Normalise: $1.mm \times 2^{exp}$

⇒ $1.01001001100110\ldots \times 2^{4}$ (1)

Stored exp = actual + excess

= 4 + 127

= 131

= 10000011 (1)

1 10000011 01001001100110011 (1)

C1A26667

(1)

Question 3

In a five stage pipeline, execution of an instruction takes 10 nsec. Suppose that the cycle time of this machine is 2 nsec, how many instructions can be executed in one second? (2 marks)

1 inst = 10 ns

1 cycle = 2 ns

X = 1 s

X = 500 MIPS

Question 4

You received an encoded Hamming code word 011001000011. You want to determine what the original data word sent was. Assume the received code word was encoded for one-bit error detection and correction using Hamming code with even parity. What was the original data word sent? Express your answer as a hexadecimal value. (3 marks)

(1) Codeword = 011001000011

1	2	3	4	5	6	7	8	9	10	11	12	~~13~~	~~14~~	~~15~~	~~16~~
0	1	1	0	0	1	0	0	0	0	1	1				

(1)

Bit 1:	1, 3, 5, 7, 9, 11	2 ✓
Bit 2:	2, 3, 6, 7, 10, 11	4 ✓
Bit 4:	4, 5, 6, 7, 12	2 ✓
Bit 8:	8, 9, 10, 11, 12, ~~13~~	2 ✓

∴ No error

(1) Data word = 10100011

⇒ A3

Question 5

A disk drive has 1000 cylinders, 8 tracks per cylinder, 32 sectors per track, and 512 bytes per sector. The rotation time is 4ms. The read/write head takes 10ms to move between adjacent cylinders and the maximum seek time is 300ms. A data file contains 200KB of data stored consecutively. What is the average time needed to read the data? (3 marks)

1

# of tracks = $200\,KB \times \frac{1024\,B}{1\,KB} \times \frac{1\,sect}{512\,B} \times \frac{1\,track}{32\,sect}$

= 12.5 tracks

transfer time = # of tracks × RT

= 12.5 × 4 ms

= 50 ms

1

track	Prob	No. of S	S/W
1	1/8	1	12
2	⋮	1	12
3	⋮	1	12
4	⋮	1	12
5	⋮	2	24
6	⋮	2	24
7	⋮	2	24
8	1/8	2	24

$4 \times \left(\frac{1}{8} \times 12\right) = 6$

$4 \times \left(\frac{1}{8} \times 24\right) = 12$

Avg s/w = 6 + 12 = 18 ms.

Question 6

a) Which levels are always interpreted in a six level computer? (1 mark)

Levels 2 & 3

1

Avg time = Avg Seek + Avg Rotation * transfert AS

= 300/3 + 4/2 + 50 + 18

= 170 ms.

b) Which levels are intended for application programmers in a six level computer? (1mark)

5, 4 $\frac{1}{2}$ mark each.

c) Why was microprogramming invented? (2 marks)

- Interpreter => to execute ISA level programs
- h/w will now have to execute microprogram instead of ISA-level programs.

Question 7

Write the output for the following machine language program if the inputs are 10, 20, 30, 45 and -1. The inputs are in sequential order. Show the accumulator value for each input and any changes to it done by the program. (5 marks)

0	0011000000001010	3	10
2	1011000000001010	11	10
4	1000000000001110	8	14
6	0010000000001110	2	14
8	1001000000000000	9	0
10	0100000000001110	4	14
12	0000000000000000	0	0
14	0000000000000001	0	1

Output = -1

Acc = ~~10~~ ~~10~~ ~~20~~ ~~2~~ ~~30~~ ~~15~~ ~~45~~ ~~3~~ -1

Question 8 (10 marks)

A function is defined as follows:

$$f(n) = \begin{cases} n^4 & for\ n < 0 \\ 25 & for\ 0 \geq n \leq 3 \\ n^3 - 4n + 5 & for\ n > 3 \end{cases}$$

$0 \leq n \leq 3$

Write Wombat1 machine language program that reads an input integer value *n*, and computes and displays the output of the above function *f(n)*. Write your program in the decimal version beginning with the memory/line address for each line.

```
0: 3 0 ; read n
2: 2 44; store n
4: 11 34; jump to loc 34 for n4 Test.
6: 5 48; Acc = Acc + -4.
8: 10 14; if Acc=0 ; jump n3-4n+5
10: 1 50; load 25
12: 9 40; jump to write
14: 1 44; load n
16: 7 54; 4n
18: 5 52; 4n+5
20: 2 46; store 4n+5
22: 1 44; load n
24: 7 44; nxn
26: 7 44; nxnxn
28: 6 46; n3-4n+5
30: 9 40; jump to write
32: 1 44; load n
34: 7 44; nxn
36: 7 44; nxnxn
38: 7 44; n4
40: 4 0 ; write.
42: 0 0 ; stop
44: 0 0 ; storage n
46: 0 0 ; storage n3
48: 0 -4 ; -4
50: 0 25 ; 25
52: 0 5 ; 5
54: 0 4 ; 4
```

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